마이크로프로세서

Chapter 0. Orientation & Development Environment

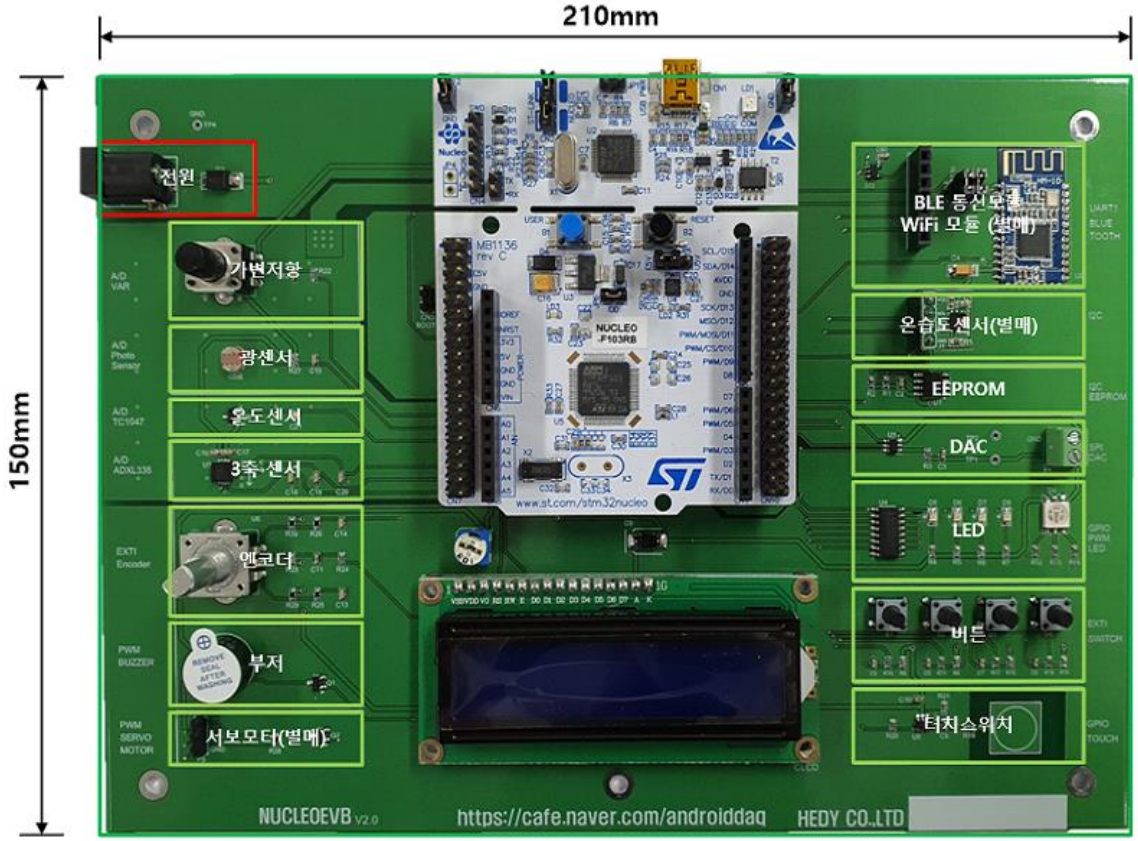
Human-Robot Interaction Laboratory

KYUNG HEE UNIVERSITY

강의 개요

>> 마이크로프로세서

- 강의 진행 방식
- 실습 키트 및 참고 도서 안내



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개발 환경

>> STM32CubeIDE & STM32CubeMX

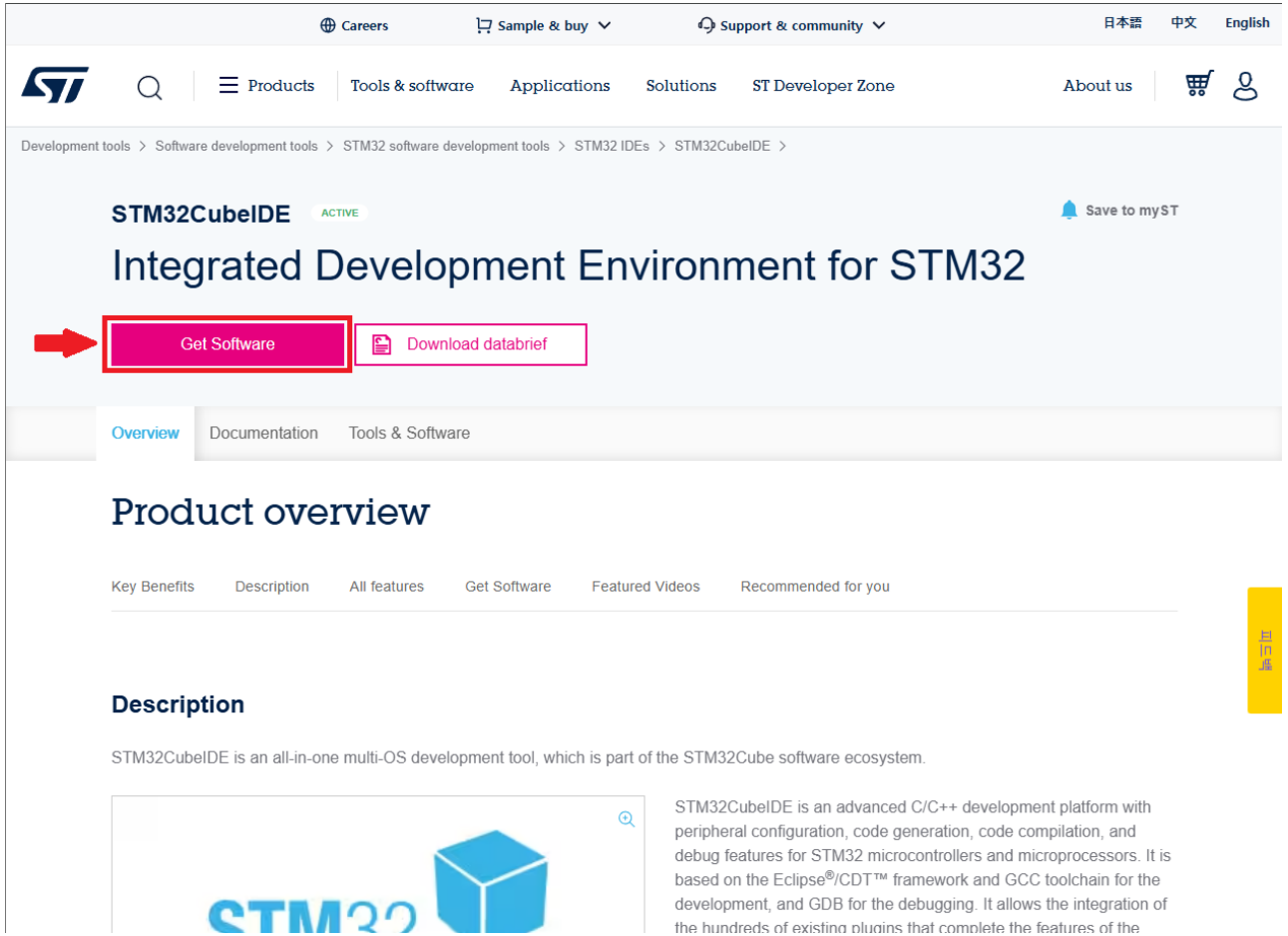
- **STM32CubeIDE**
  - STM32 마이크로컨트롤러 및 마이크로프로세서 개발을 위한 C/C++ 개발 플랫폼
  - Eclipse 기반의 GCC툴체인, GDB 디버깅 기능 제공
  - STM32CubeMX와의 연동을 통해 프로젝트 관리 및 코드 생성, 재생성
  - 다양한 확장 패키지 지원

- **STM32CubeMX**
  - STM32 마이크로컨트롤러 및 마이크로프로세서 구성을 돕는 도구
  - Cortax-M 시리즈의 C 코드 생성
  - 핀 배열, 통신 설정 등 다양한 하드웨어 기능을 GUI를 통해 직관적으로 설정 가능
  - 클럭 설정 , 하드웨어 설정, 핀 충돌 검증 등 다양한 기능 제공

개발 환경

>> STM32CubeIDE

- **STM32CubeIDE 설치**
  https://www.st.com/en/development-tools/stm32cubeide.html



개발 환경

>> STM32CubeIDE

- **STM32CubeIDE 설치**
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Get Software

	Part Number	General Description	Latest version	Download	All versions
+	STM32CubeIDE-DEB	STM32CubeIDE Debian Linux Installer	1.17.0	Get latest	Select version
+	STM32CubeIDE-Lnx	STM32CubeIDE Generic Linux Installer	1.17.0	Get latest	Select version
+	STM32CubeIDE-Mac	STM32CubeIDE macOS Installer	1.17.0	Get latest	Select version
+	STM32CubeIDE-RPM	STM32CubeIDE RPM Linux Installer	1.17.0	Get latest	Select version
+	STM32CubeIDE-Win	STM32CubeIDE Windows Installer	1.17.0	Get latest	Select version

STM32 CubeIDE

Looking for previous STM32CubeIDE versions?

Please open a case to our support team. We're here to assist you!

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>> STM32CubeIDE

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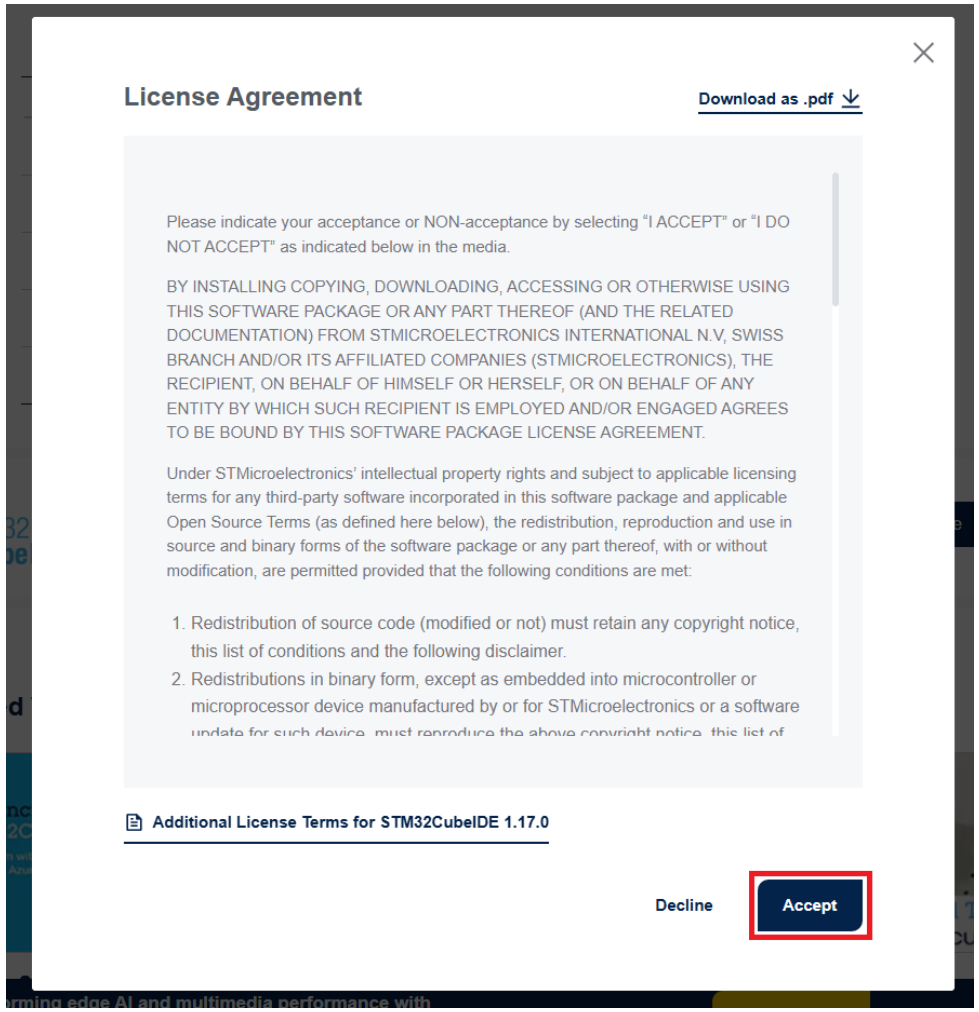

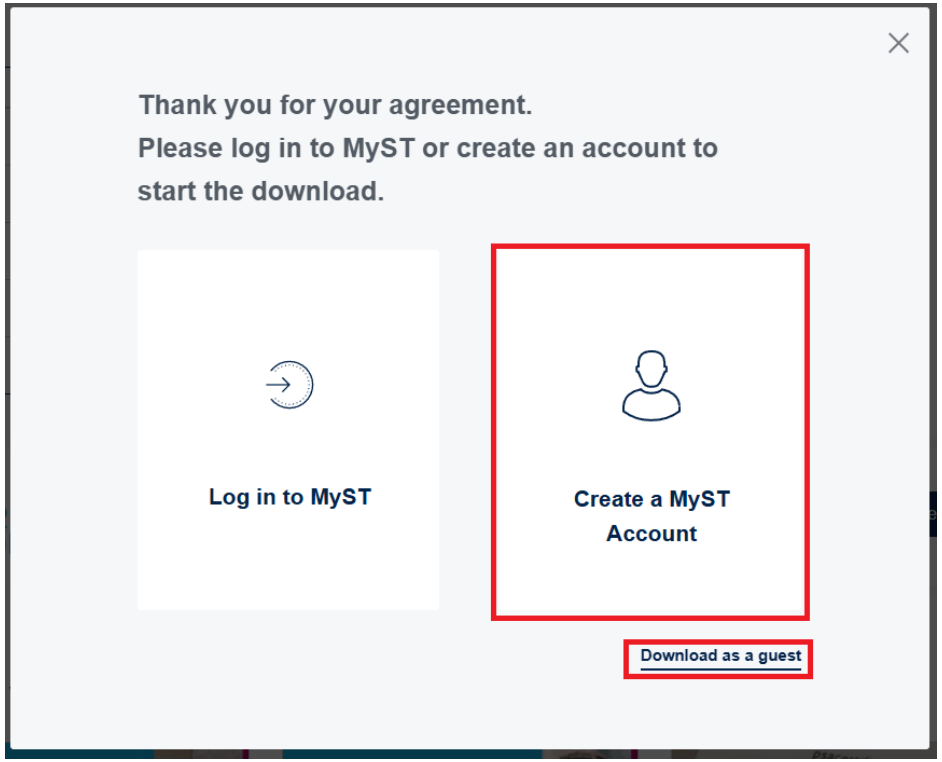

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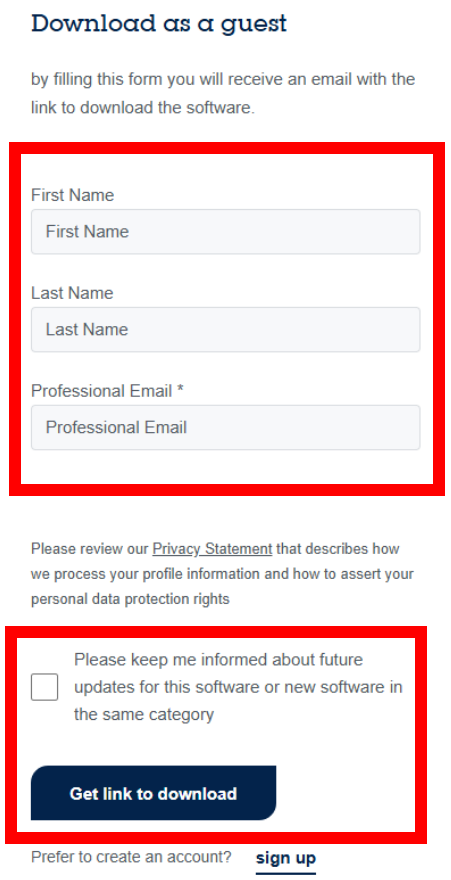

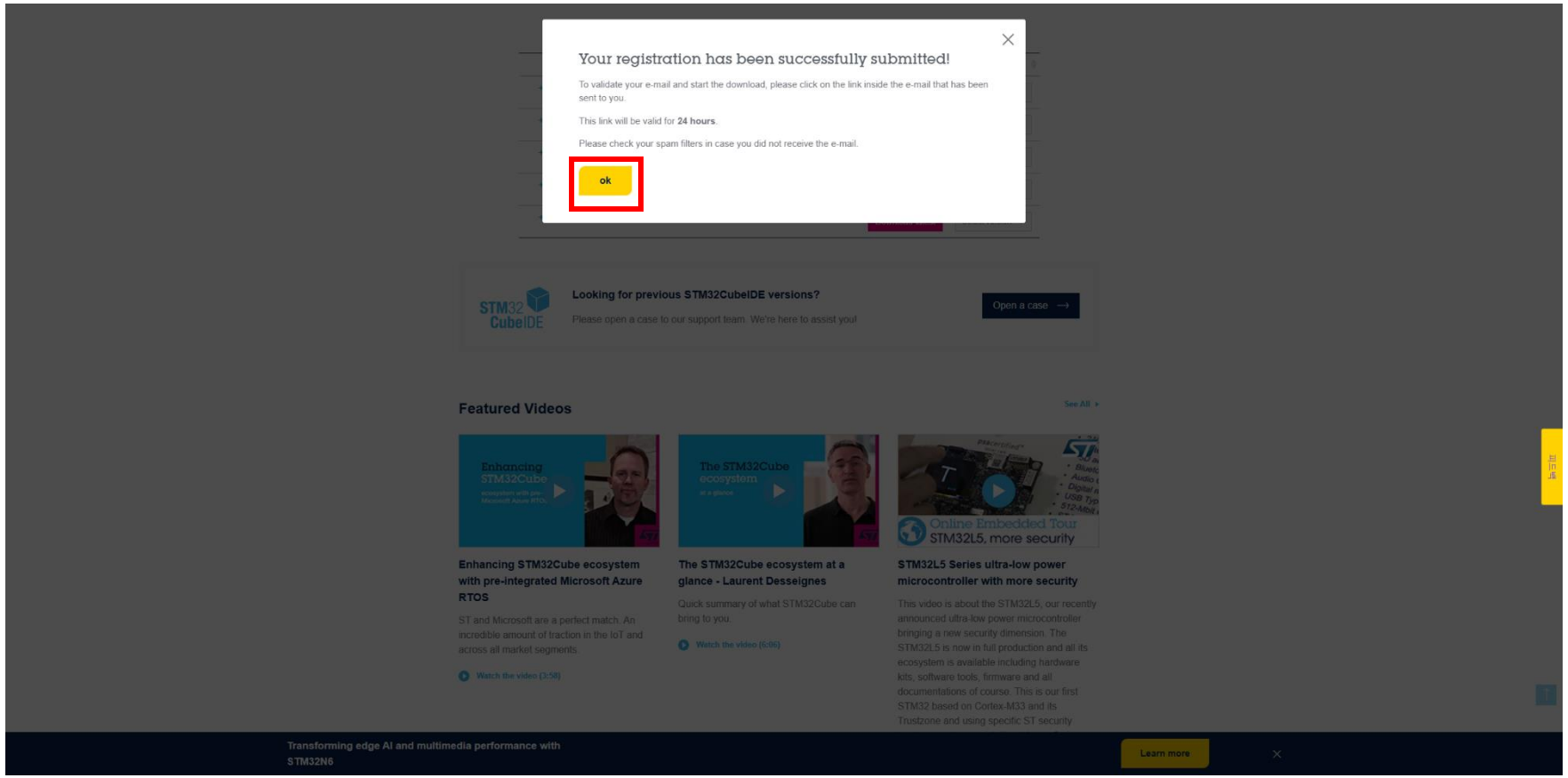

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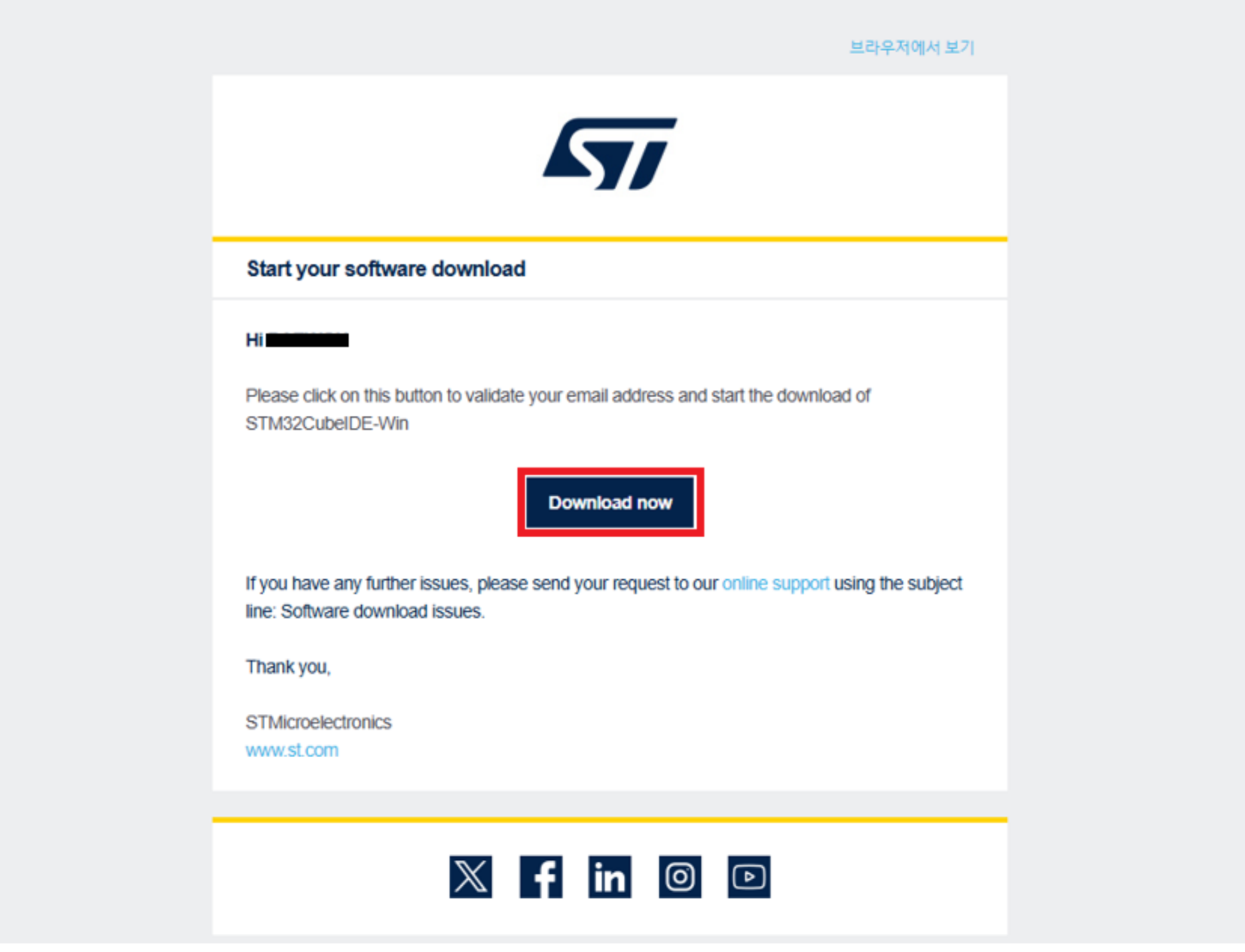

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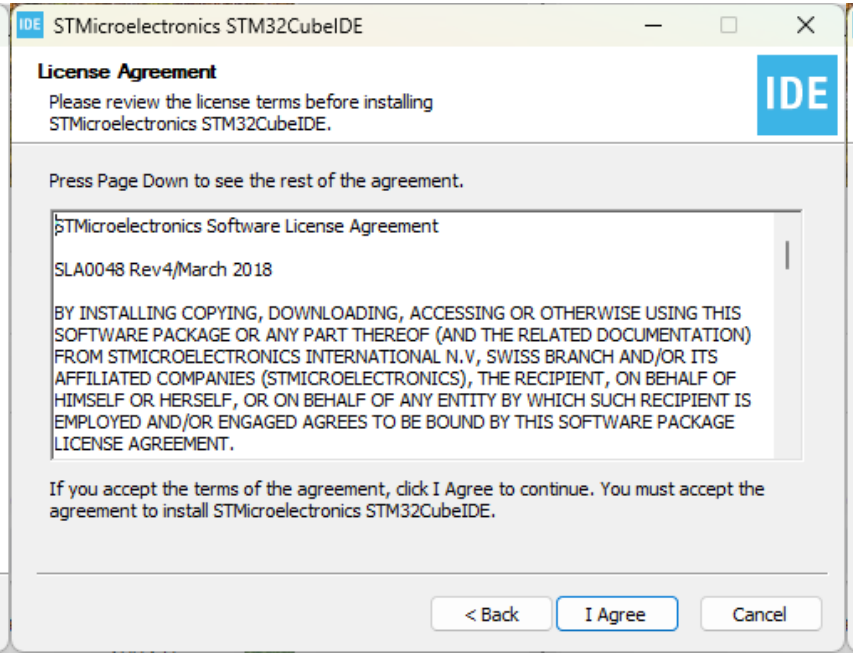

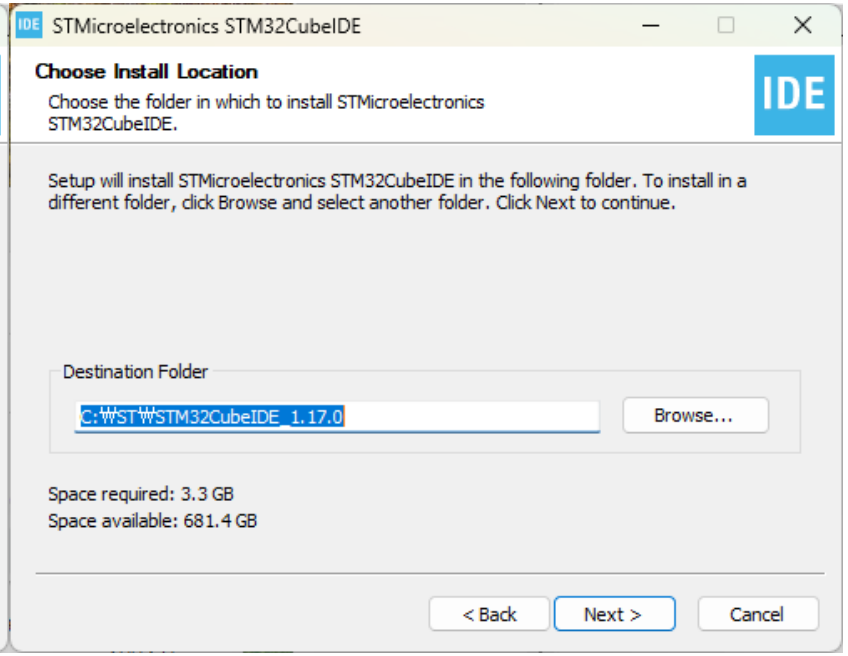

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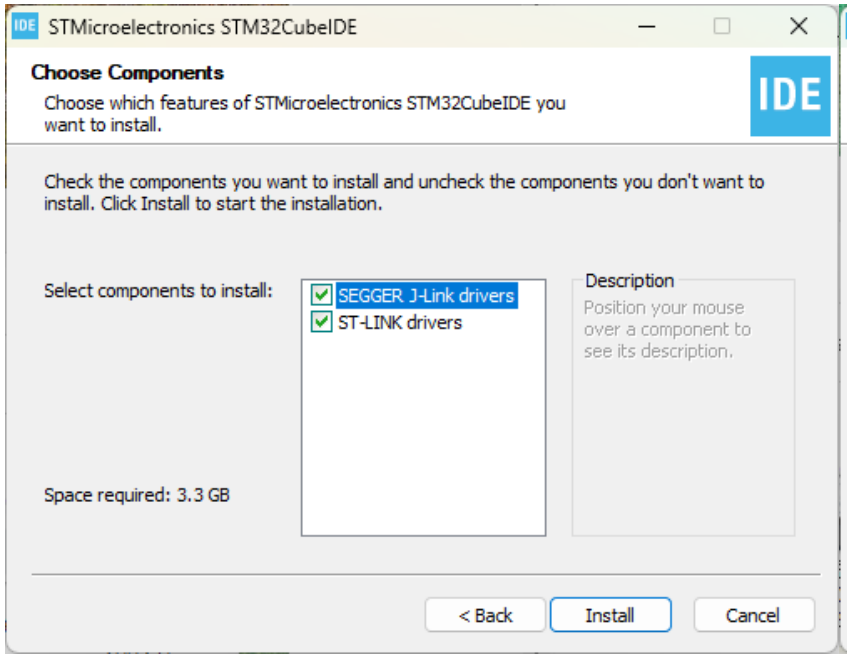
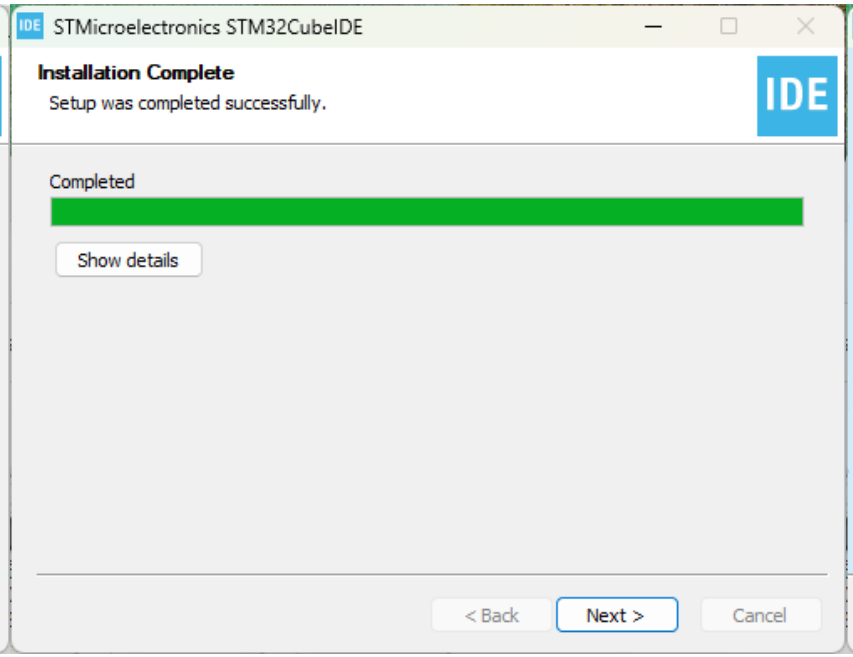
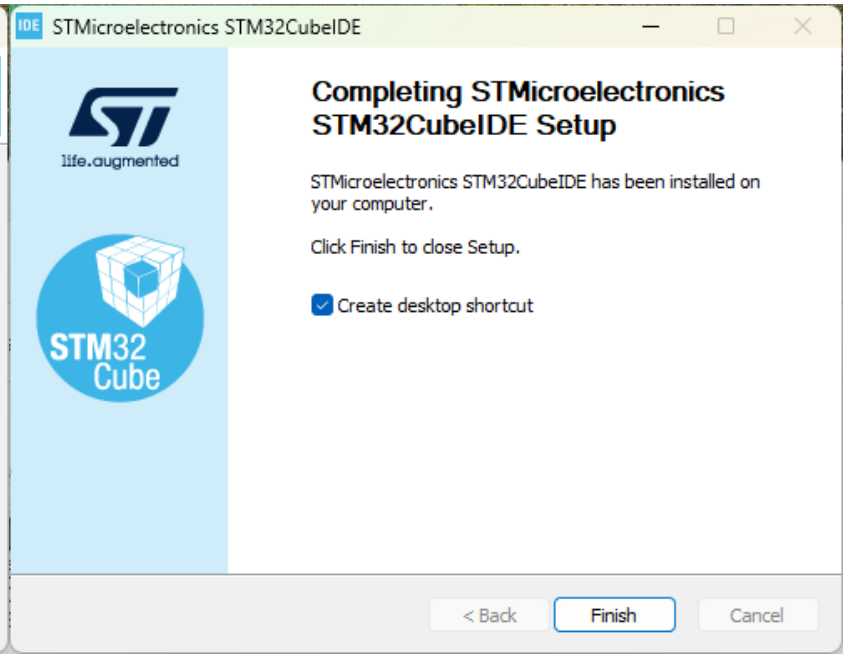


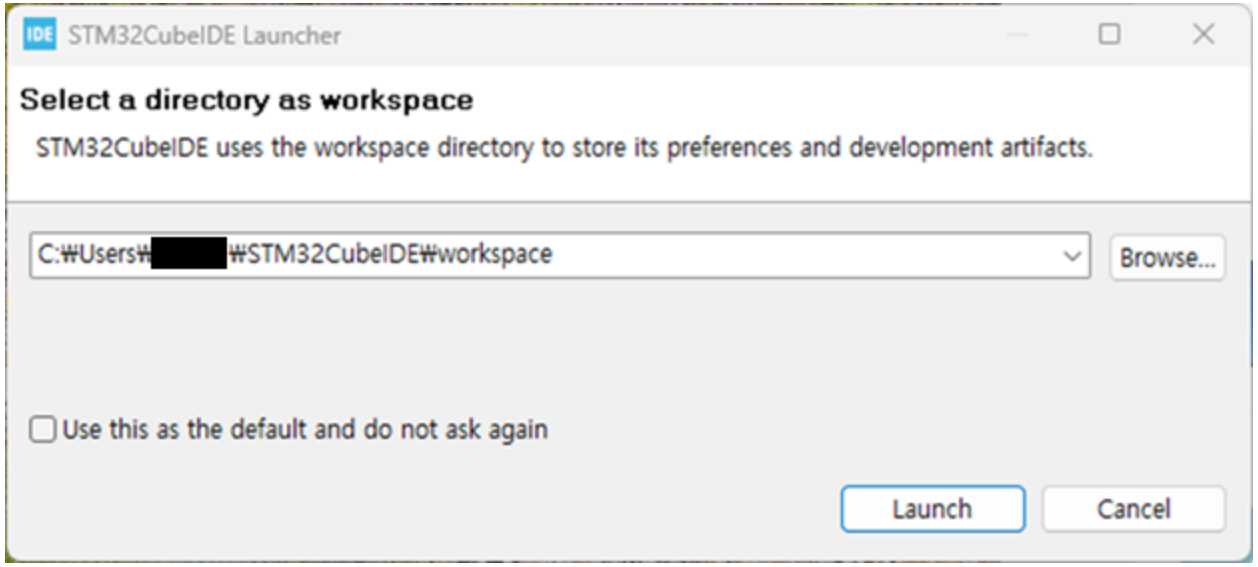




개발 환경

>> STM32CubeIDE

- STM32CubeIDE 초기설정



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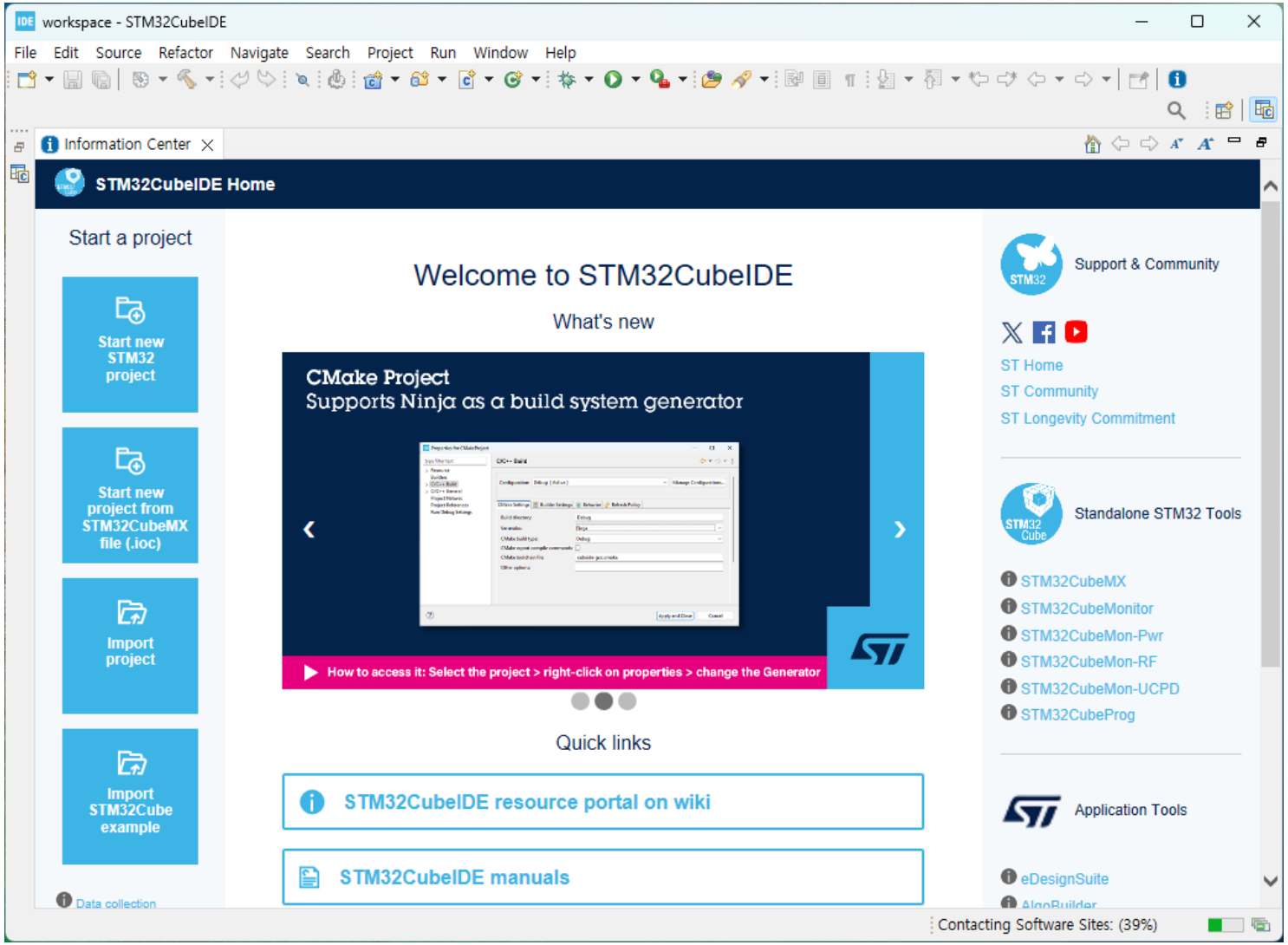
>> STM32CubeIDE

- STM32CubeIDE 초기 설정



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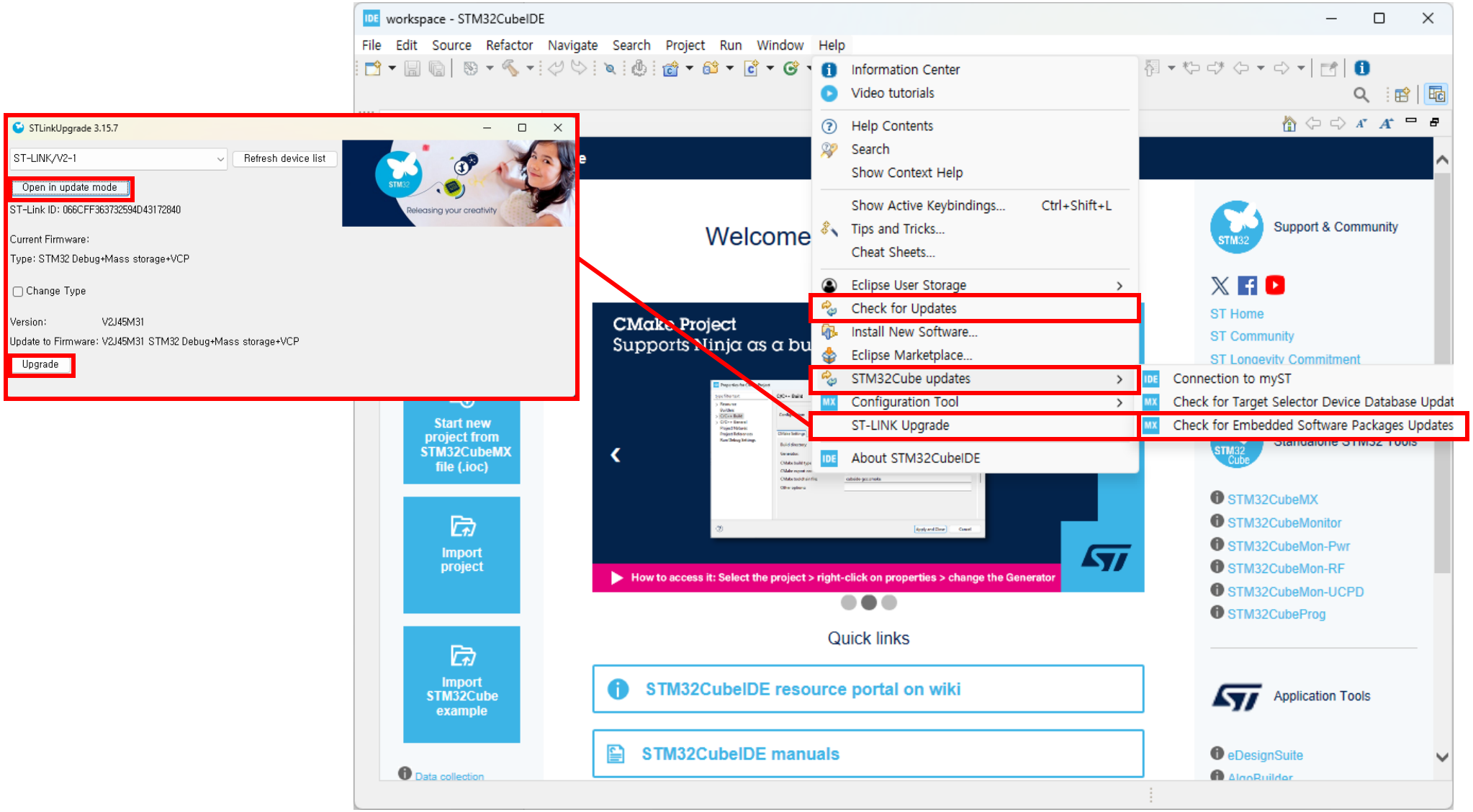

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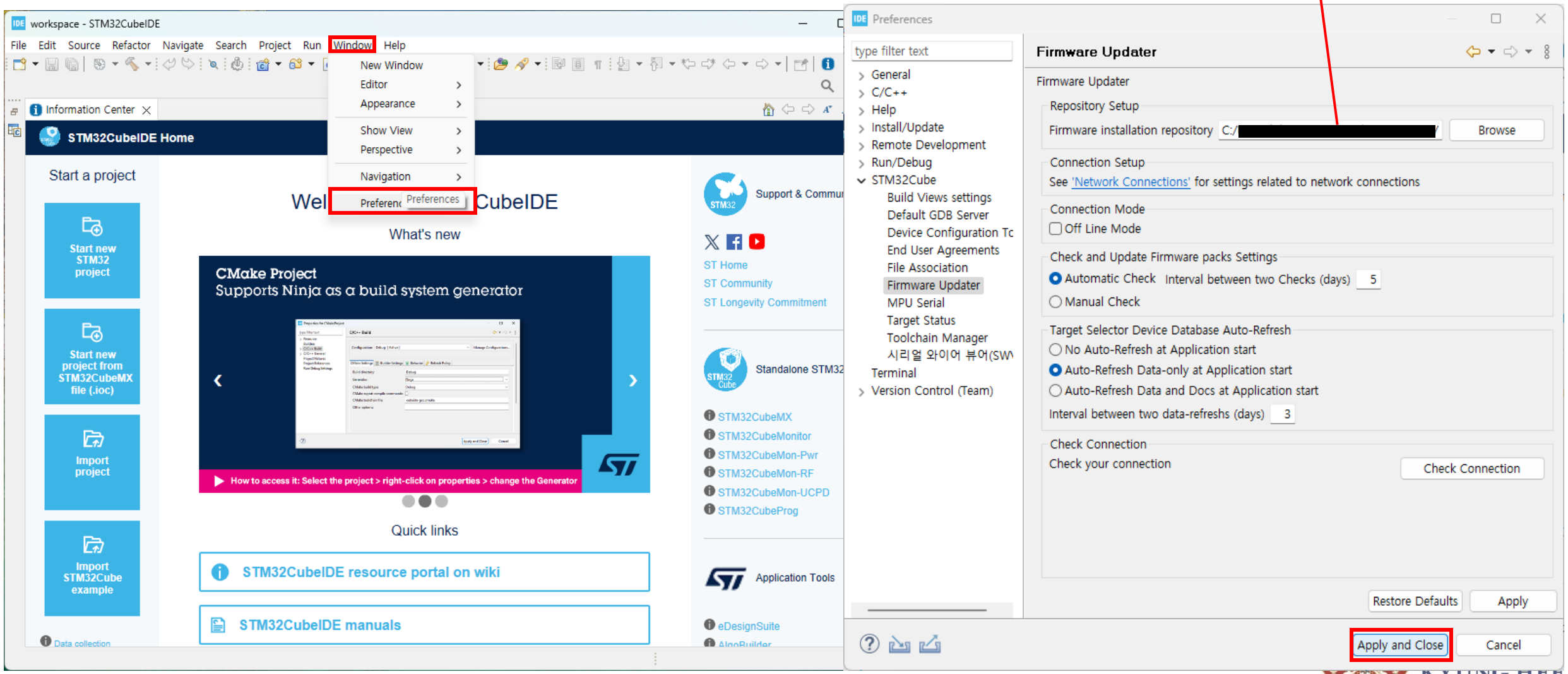
>> STM32CubeIDE

- STM32CubeIDE 초기 설정

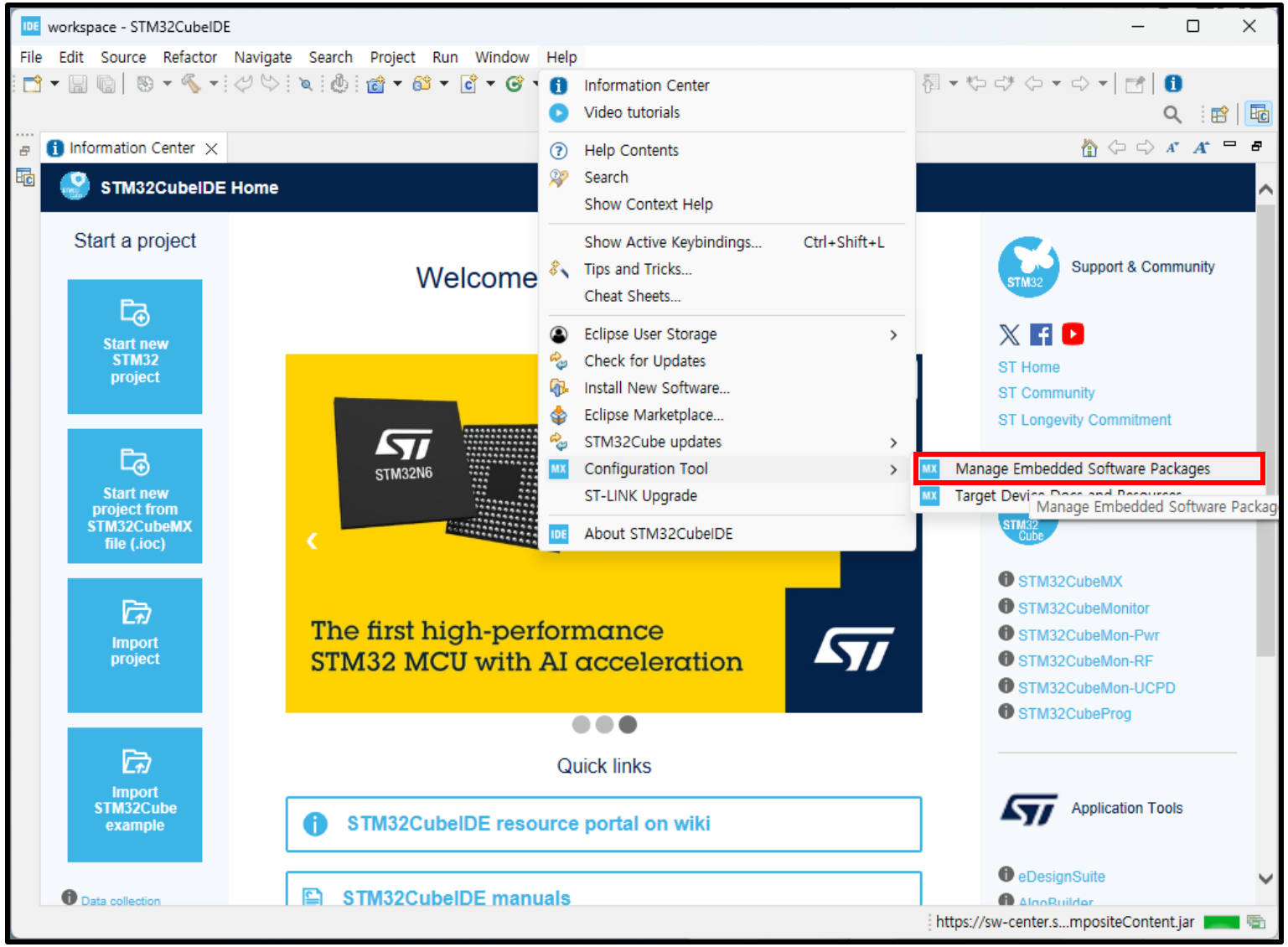
프로젝트 관리 폴더
경로 지정(선택 사항)



개발 환경

>> STM32CubeIDE

- STM32CubeIDE 초기 설정



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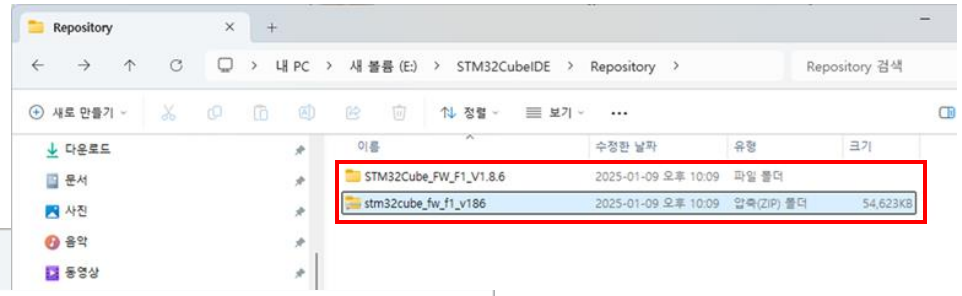
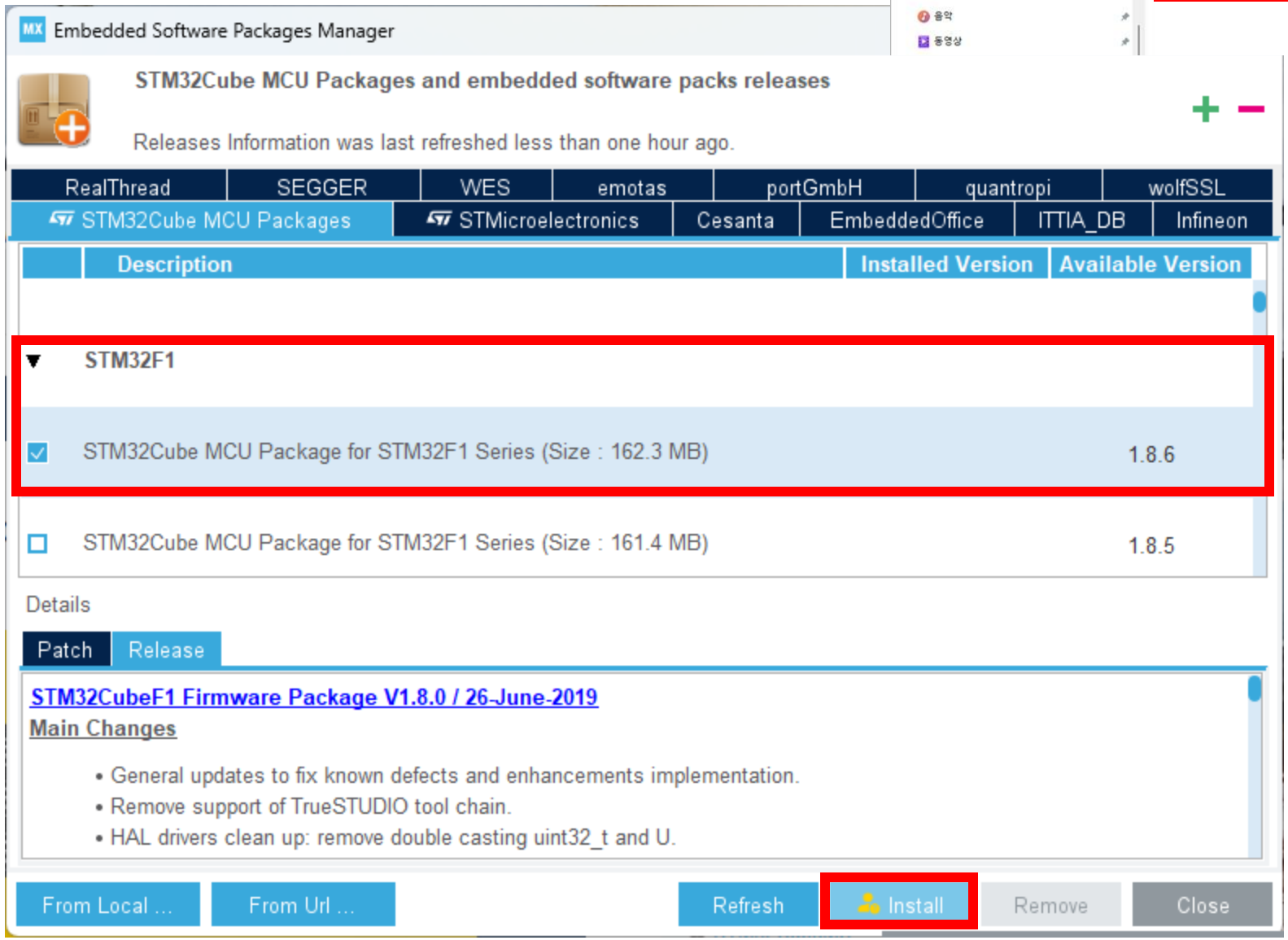
- STM32CubeIDE 초기 설정





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>> STM32Cube

- STM32CubeMX 설치
  https://www.st.com/en/development-tools/stm32cubemx.html

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Product overview

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Description

STM32CubeMX is a graphical tool that allows a very easy configuration of STM32 microcontrollers and microprocessors, as well as the gener
the corresponding initialization C code for the Arm® Cortex®-M core or a partial Linux® Device Tree for Arm® Cortex®-A core, through a step-b
process.

The first step consists in selecting either an STMicrolectronics STM32 microcontroller, microprocessor or a development platform that matches the required set of peripherals, or an example running on a specific development platform.

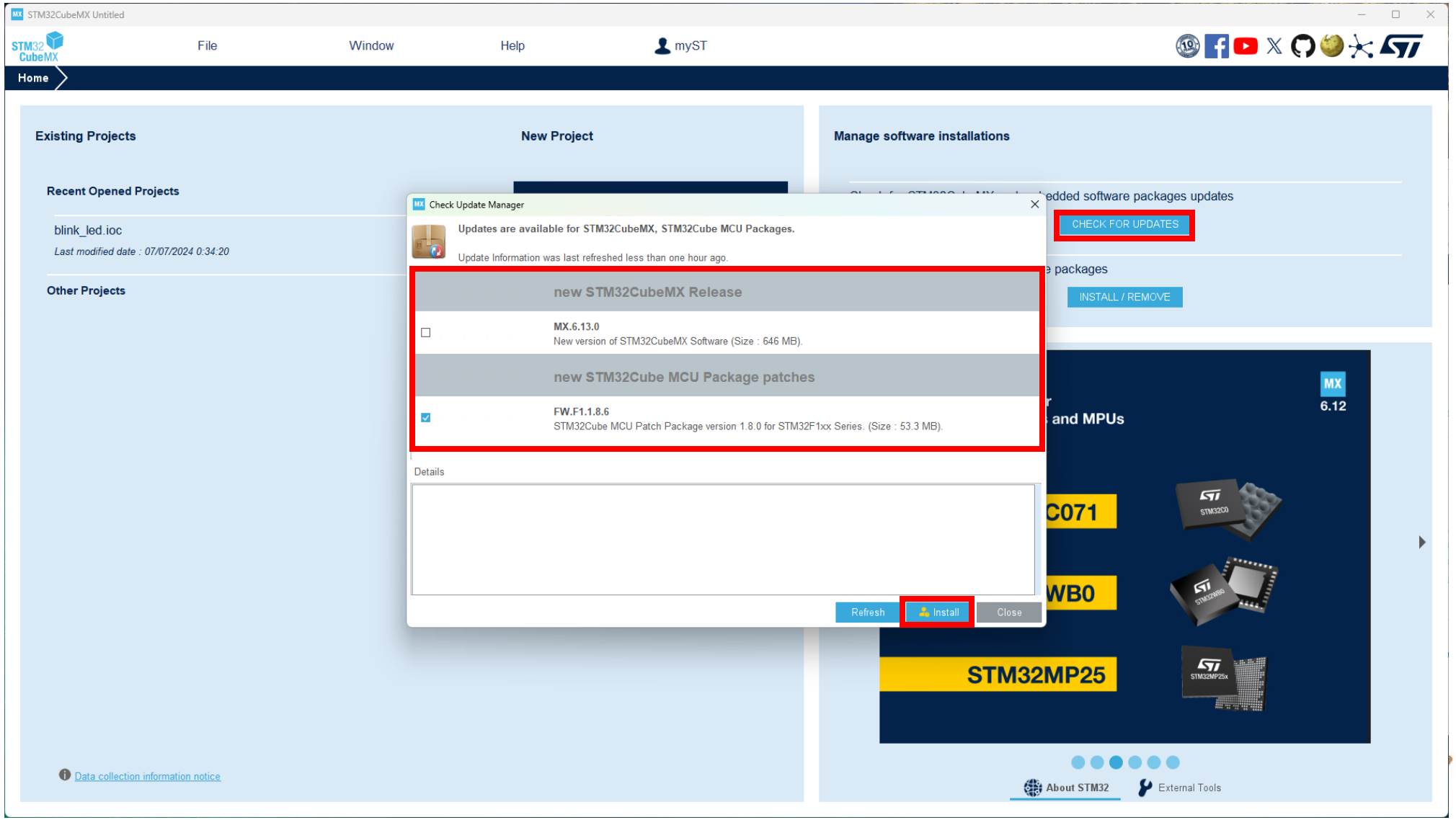
Get Software

	Part Number	General Description	Latest version	Download	All versions
+	STM32CubeMX-Lin	STM32Cube init code generator for Linux	6.13.0	Get latest	Select version
+	STM32CubeMX-Mac	STM32Cube init code generator for macOS	6.13.0	Get latest	Select version
+	STM32CubeMX-Win	STM32Cube init code generator for Windows	6.13.0	Get latest	Select version

개발 환경

>> STM32CubeIDE

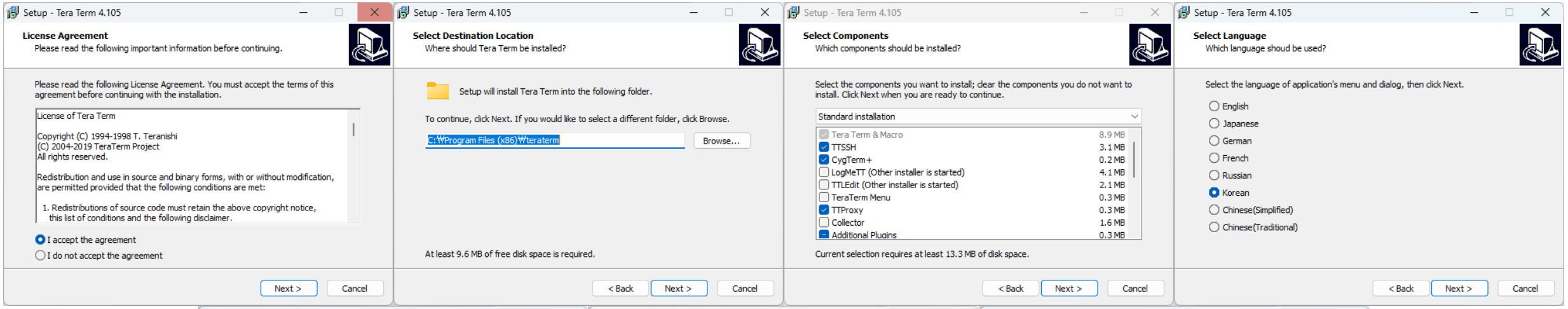
- STM32CubeMX 설치



개발 환경

>> Tera Turm

- Tera Turm 설치 (Putty 대체 가능)
  https://teratermproject.github.io/index-en.html



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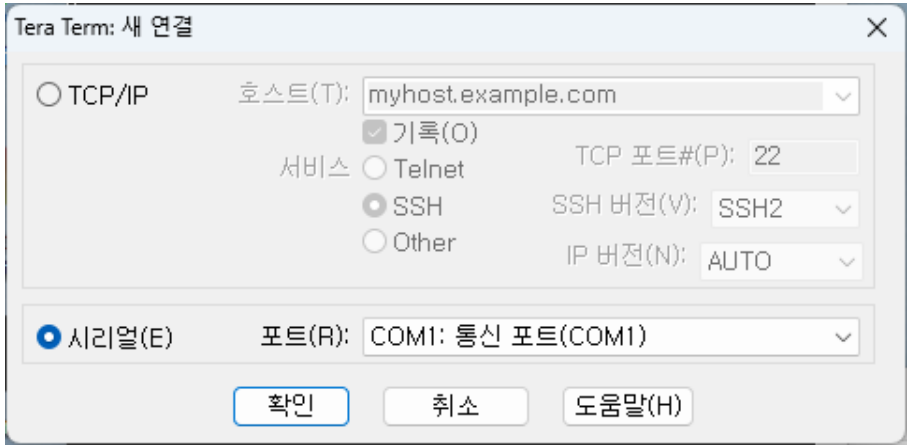
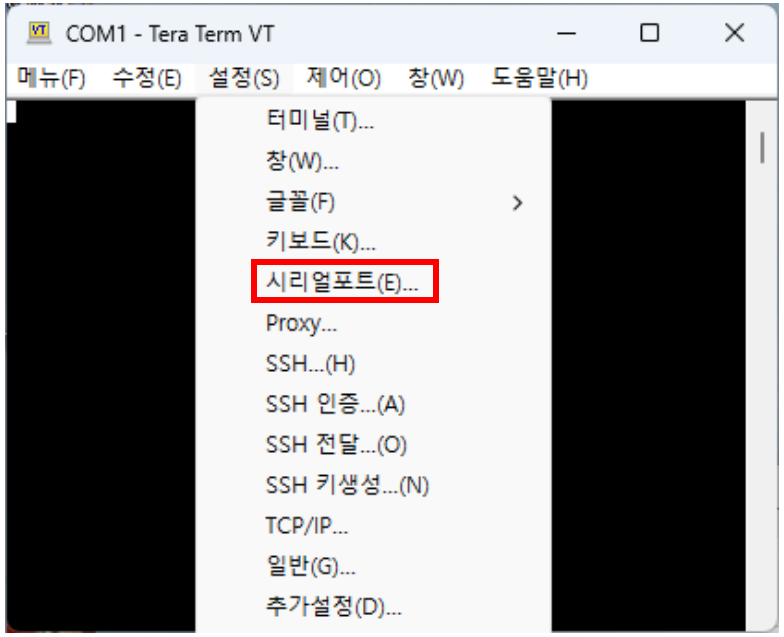
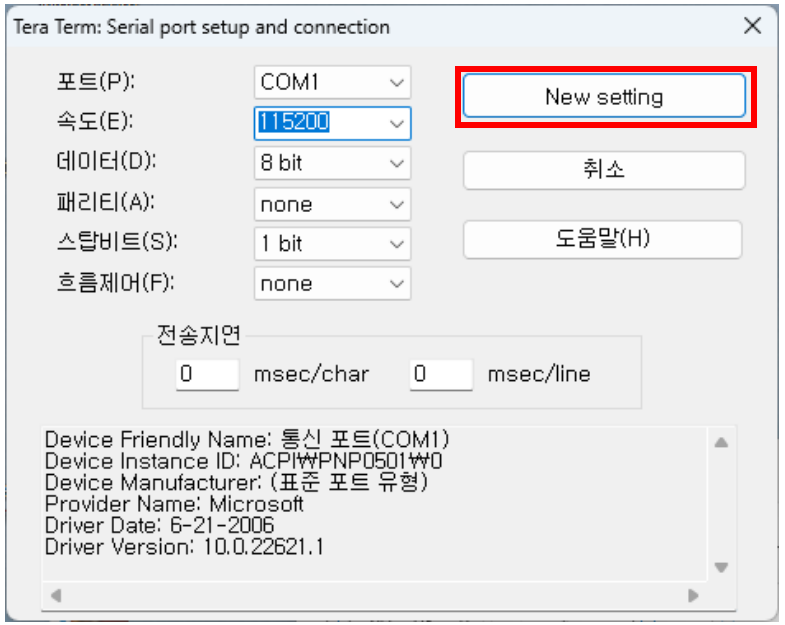
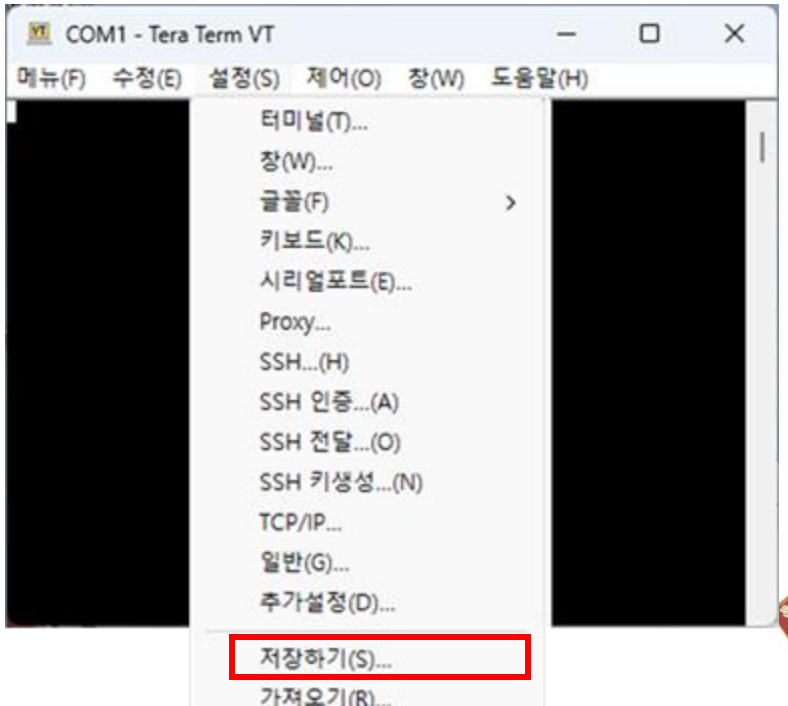
개발 환경

>> Tera Turm









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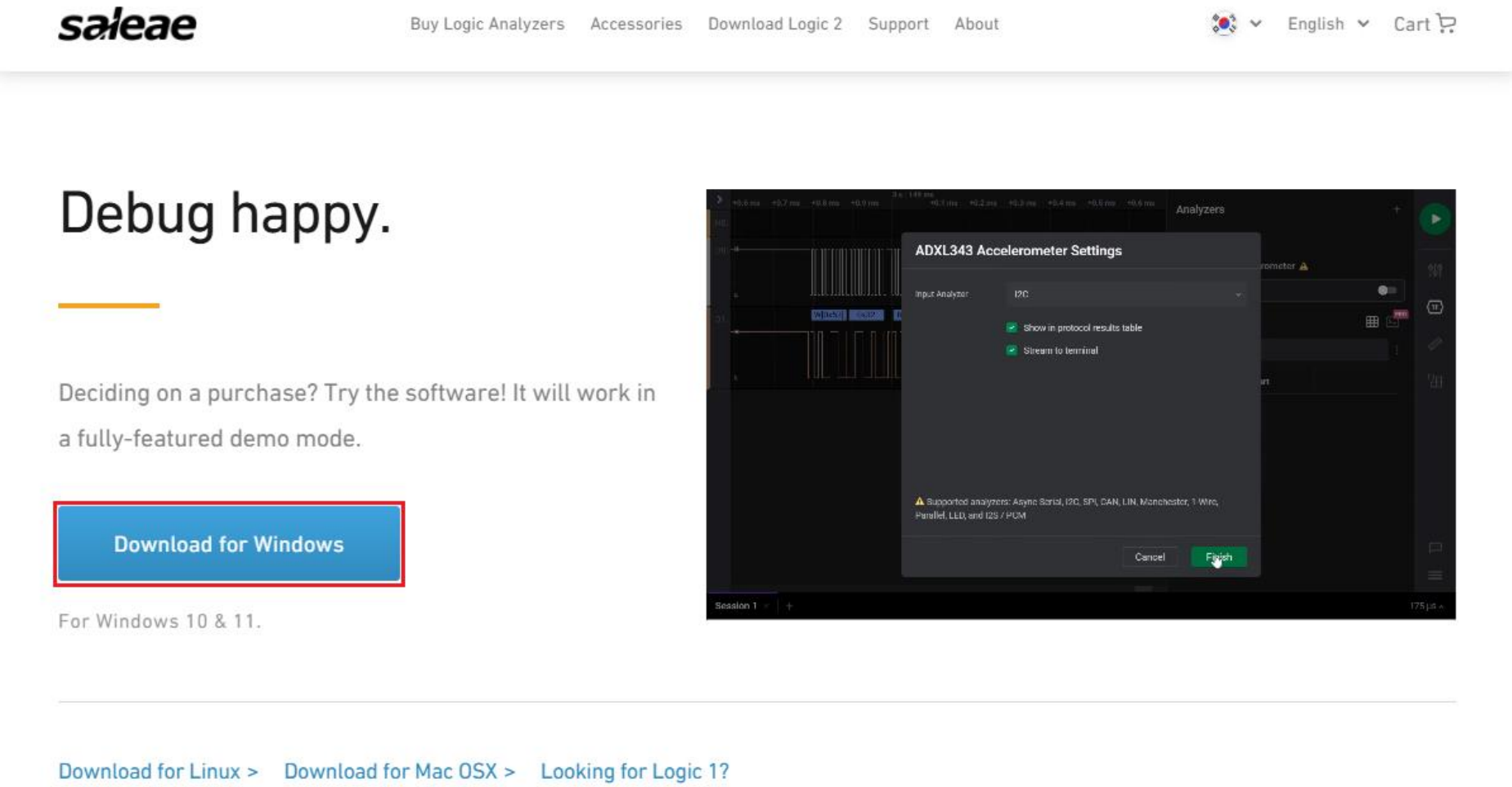
개발 환경

>> Logic 2

- Logic 2
  https://www.saleae.com/pages/downloads



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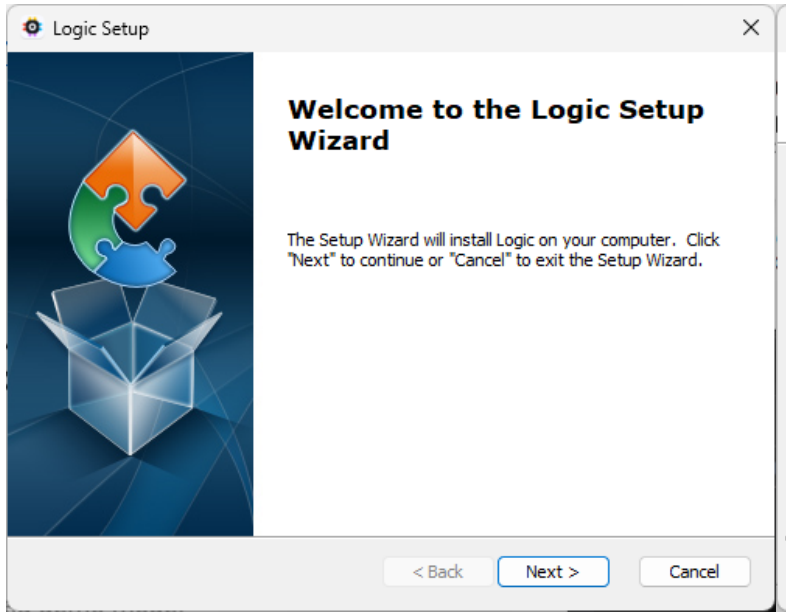
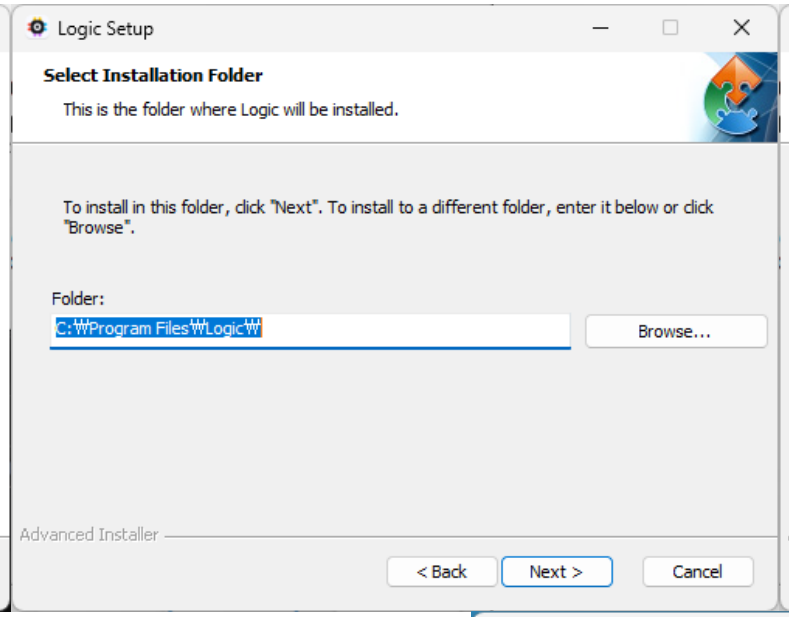
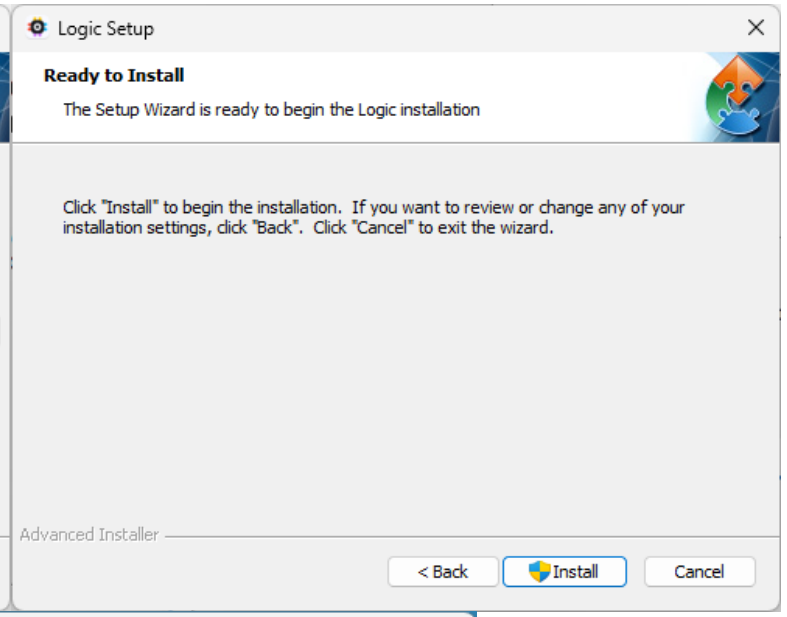
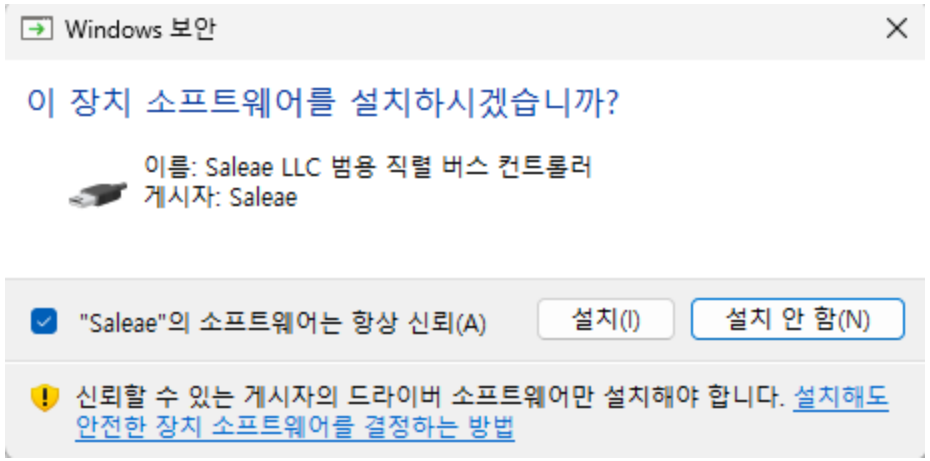
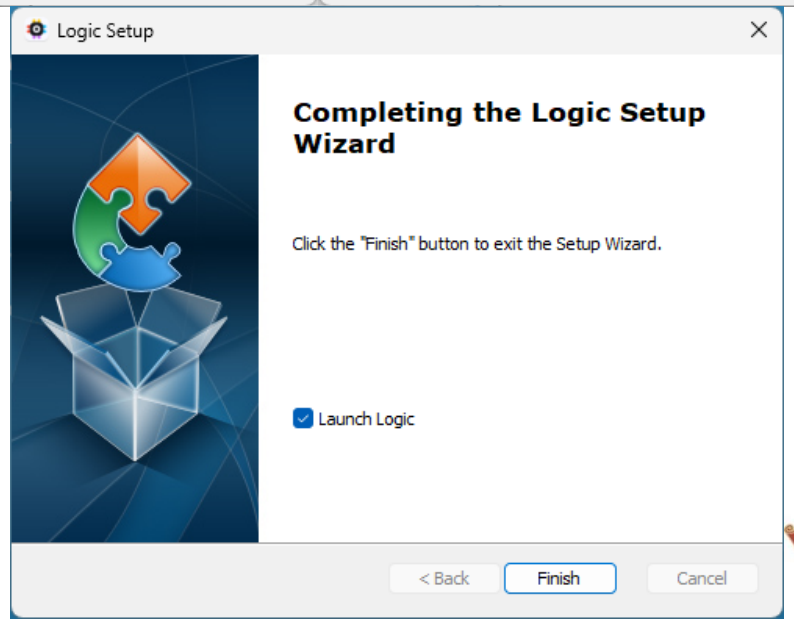
>> Logic 2











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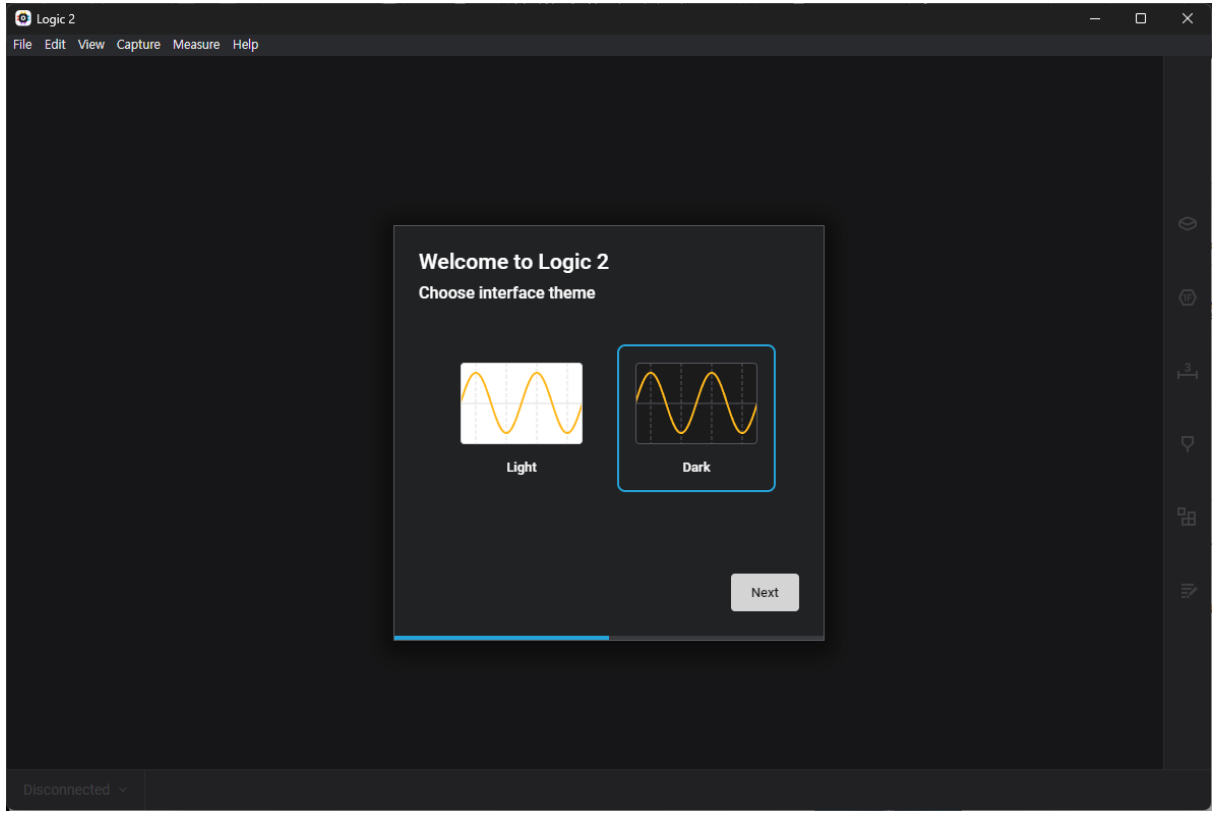
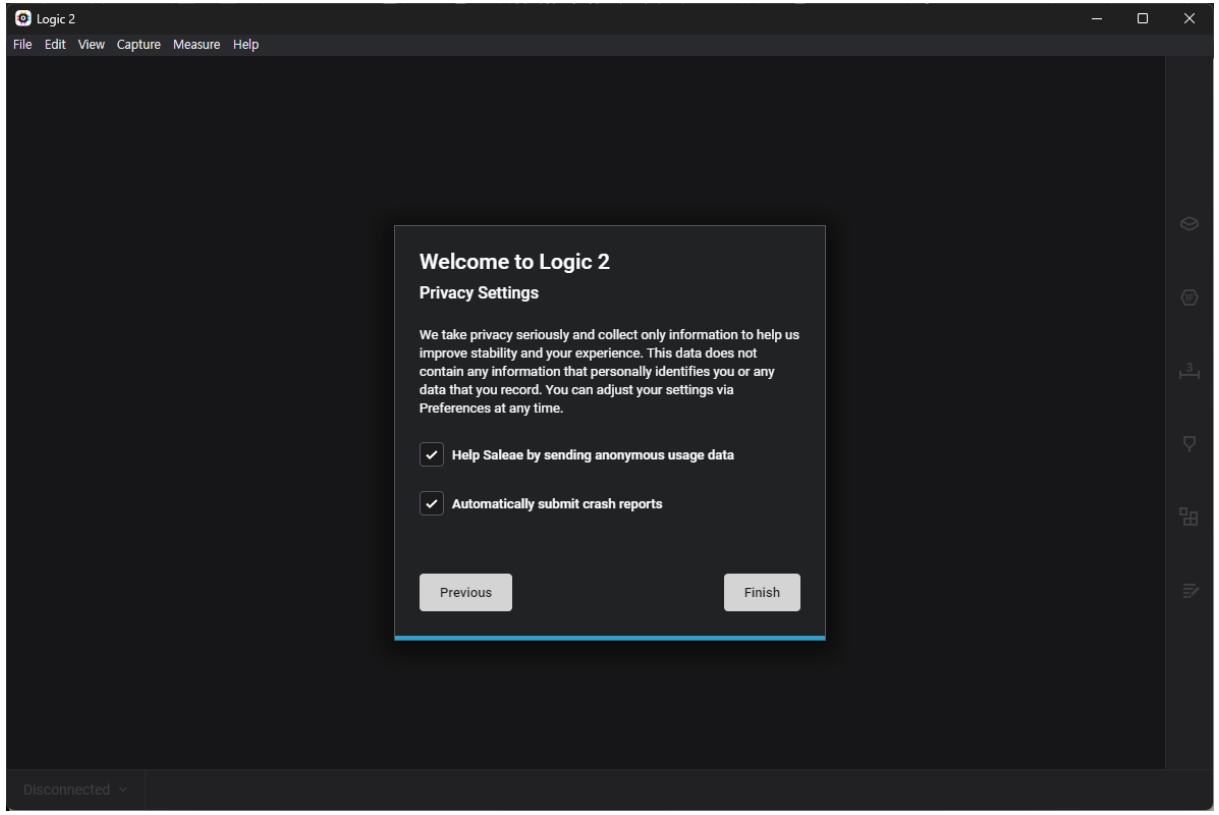
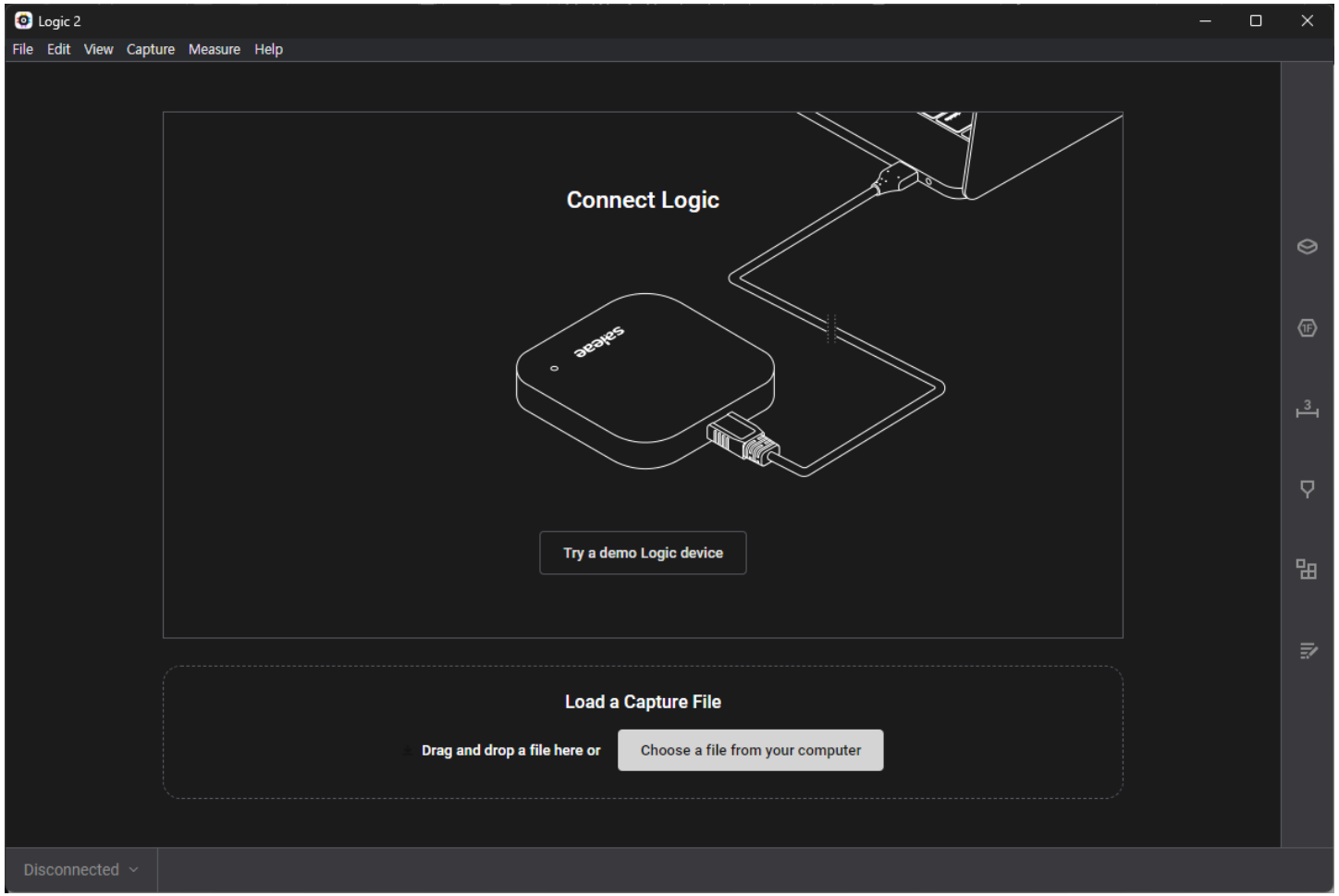
>> Logic 2


Logic 2
File Edit View Capture Measure Help
Welcome to Logic 2
Choose interface theme
Light
Dark
Next
Disconnected


Logic 2
File Edit View Capture Measure Help
Welcome to Logic 2
Privacy Settings
We take privacy seriously and collect only information to help us improve stability and your experience. This data does not contain any information that personally identifies you or any data that you record. You can adjust your settings via Preferences at any time.
Help Saleae by sending anonymous usage data
Automatically submit crash reports
Previous
Finish
Disconnected

개발 환경

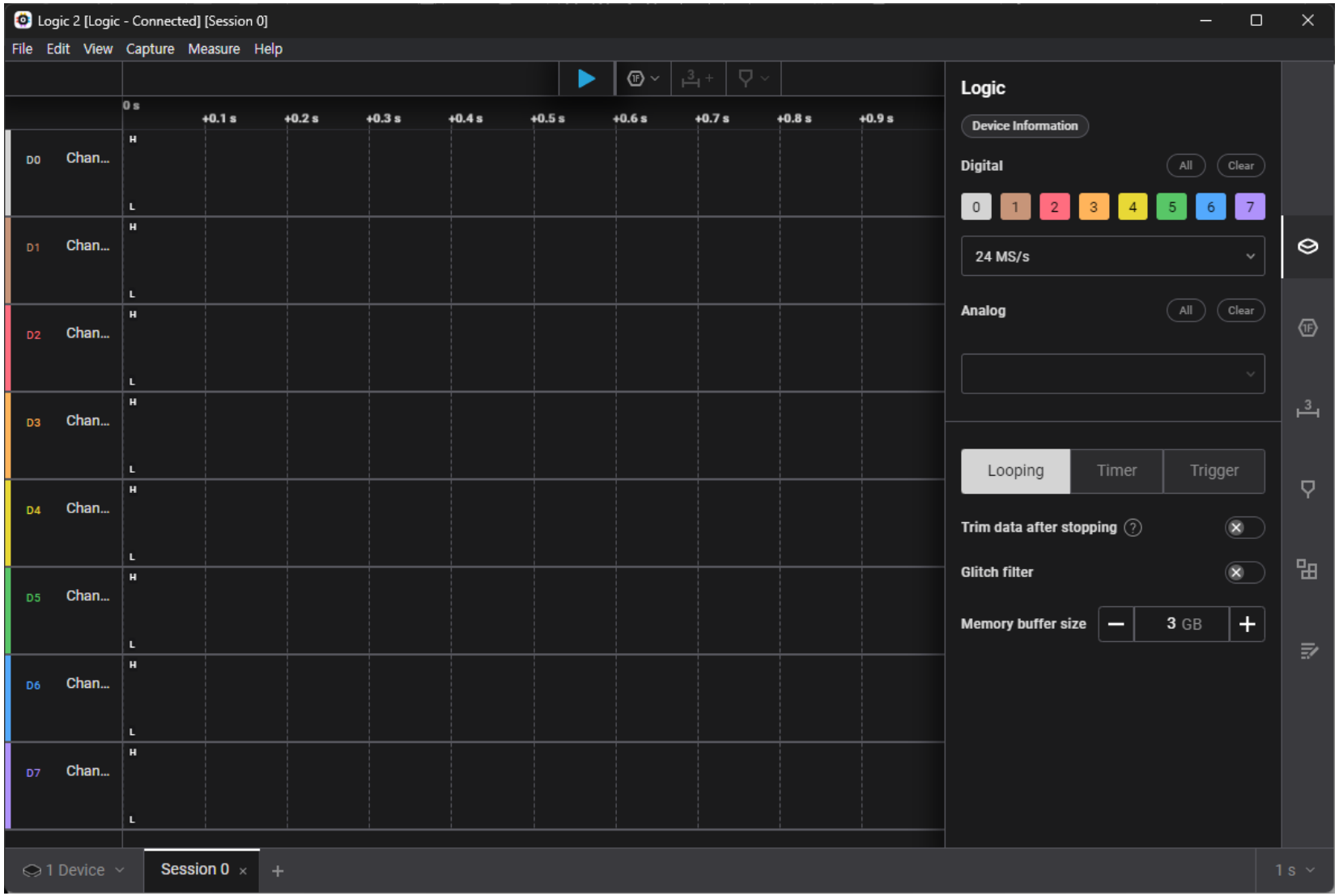
>> Logic 2


Logic 2
File Edit View Capture Measure Help
Connect Logic
Try a demo Logic device
Load a Capture File
Drag and drop a file here or
Choose a file from your computer
Disconnected

24MHz8CH
CH1
CH3
CH5
CH7
CLK
CH2
CH4
CH6
CH8
GND
PWR
CH1

개발 환경

>> Logic 2


Logic 2 [Logic - Connected] [Session 0]
File Edit View Capture Measure Help
0 s
+0.1 s
+0.2 s
+0.3 s
+0.4 s
+0.5 s
+0.6 s
+0.7 s
+0.8 s
+0.9 s
D0 Chan...
D1 Chan...
D2 Chan...
D3 Chan...
D4 Chan...
D5 Chan...
D6 Chan...
D7 Chan...
H
L
Logic
Device Information
Digital
All
Clear
0 1 2 3 4 5 6 7
24 MS/s
Analog
All
Clear
Looping
Timer
Trigger
Trim data after stopping
Glitch filter
Memory buffer size
3 GB
1 Device
Session 0
1 s

Thank you

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Giheung-gu, Yongin-si, Gyeonggi-Do, 17104,
Republic of Korea
Web: https://khu-hri.weebly.com/

Human-Robot Interaction Laboratory

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